

FINAL EXAM SOLUTIONS

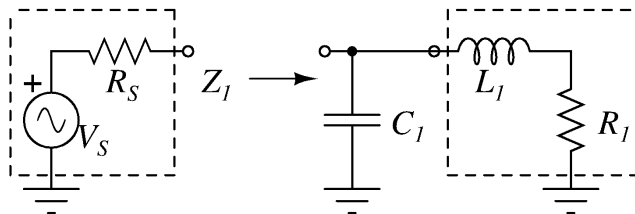
Question 1 [25 Points]:

We have a source at 27 MHz with an open-circuit output voltage (V_S) of 8 V peak-to-peak shown in dashed lines. The source resistance is $R_S=50\ \Omega$. The load is $R_L=5\ \Omega$ in series with an inductance of $L_L=118\ \text{nH}$ also shown in dashed lines.

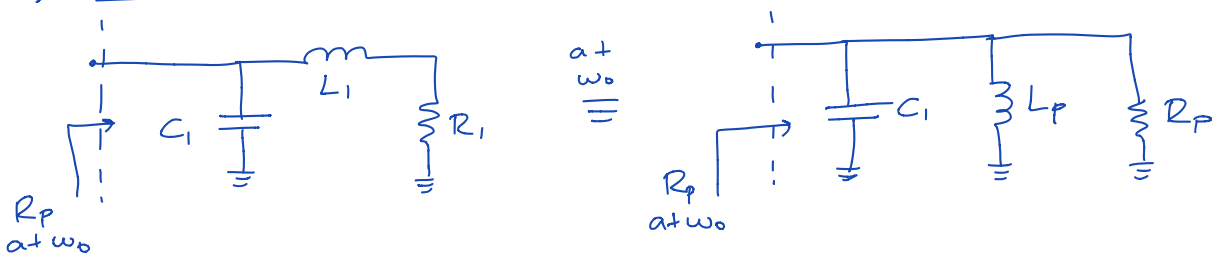
(a) Find the value of parallel capacitance C_1 to make Z_1 purely real at 27 MHz.

Important Note: Do not add any other components.

(b) Find the power delivered to R_L in mW, when this circuit is connected to the source.



a) First method: L-section transform L_L to some R_p .



$$Q = \frac{\omega_0 L_L}{R_L} = \frac{2\pi \cdot 27 \cdot 10^6 \cdot 118 \cdot 10^{-9}}{5} = 4$$

$$L_p = L_L \left(1 + \frac{1}{Q^2}\right) = 118 \cdot \left(1 + \frac{1}{4^2}\right) = 125.4\ \text{nH}$$

$$C_1 = \frac{1}{\omega_0^2 L_p} = \frac{1}{(2\pi \cdot 27 \cdot 10^6)^2 \cdot 125.4 \cdot 10^{-9}} = \boxed{277\ \text{pF}}$$

$$R_p = R_L (1 + Q^2) = 5 \cdot (1 + 4^2) = 85\ \Omega$$

Second method: Direct solution

$$\begin{aligned} Z_1 &= \frac{1}{j\omega_0 C_1} \parallel (j\omega_0 L_L + R_L) = \frac{\frac{1}{j\omega_0 C_1} (j\omega_0 L_L + R_L)}{\frac{1}{j\omega_0 C_1} + j\omega_0 L_L + R_L} \\ &= \frac{R_L + j\omega_0 L_L}{(1 - \omega_0^2 L_L C_1) + j\omega_0 R_L C_1} = \frac{a \cdot e^{j\phi_1}}{b \cdot e^{j\phi_2}} = \frac{a}{b} \cdot e^{j(\phi_1 - \phi_2)} \end{aligned}$$

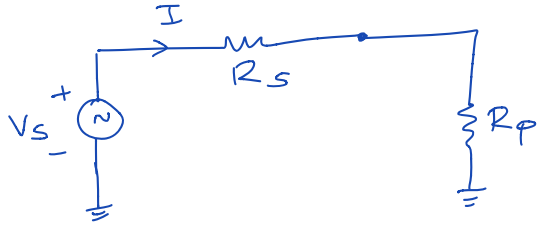
To make Z_1 real-valued, numerator and denominator should have the same phase, i.e., $\phi_1 = \phi_2$:

$$\frac{\omega_0 L_L}{R_L} = \frac{\omega_0 R_L C_1}{1 - \omega_0^2 L_L C_1}$$

$$C_1 = \frac{L_L}{\omega_0^2 L_L^2 + R_L^2} = \frac{118 \cdot 10^{-9}}{(2\pi \cdot 27 \cdot 10^6 \cdot 118 \cdot 10^{-9})^2 + 5^2} = \boxed{277\ \text{pF}}$$

Then, $Z_1 = \frac{R_L}{1 - \omega_0^2 L_L C_1} = \frac{\omega_0 L_L}{\omega_0 R_L C_1} = 85\ \Omega$ ($Q_1 = Q_2$, so ratio of real parts is the same as ratio of imaginary parts)

b) Circuit seen by source at ω_0 :



$$P_{out} = \frac{1}{2} I^2 \cdot R_p$$

$$= \frac{1}{2} \left(\frac{V_s}{R_s + R_p} \right)^2 \cdot R_p$$

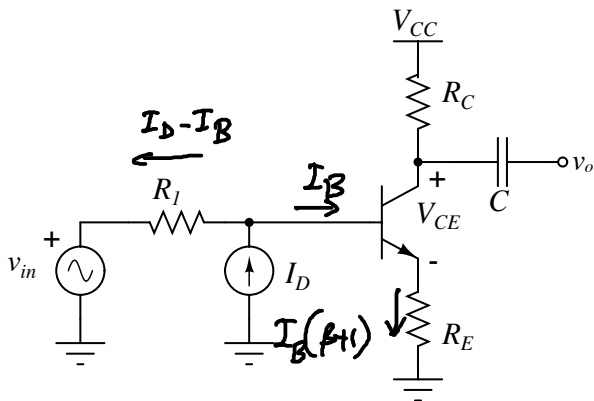
$$= \frac{1}{2} \left(\frac{8/2}{50 + 85} \right)^2 \cdot 85 = \boxed{37.3 \text{ mW}}$$

Question 2 [25 Points]:

Consider the BJT amplifier circuit shown below with $R_1=2.2\text{ K}\Omega$, $R_E=22\text{ }\Omega$, $R_C=1\text{ K}\Omega$, $I_D=0.5\text{ mA}$, $V_{CC}=12\text{ V}$. BJT parameters are $\beta=85$, $V_o=0.7\text{ V}$, $V_{SAT}=0.2\text{ V}$. C is a DC-block capacitor with a sufficiently high value.

(a) Perform DC analysis and show that the transistor is in active region by finding V_{CE} .

(b) Determine the AC small-signal voltage gain v_o/v_{in} .

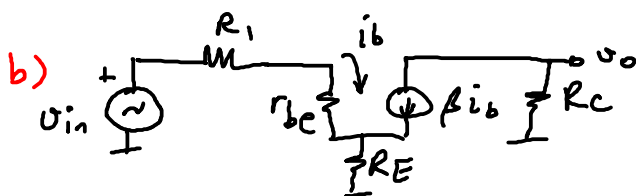


a) $V_o + I_B (\beta+1) R_E = (I_D - I_B) R_1$

$$I_B = \frac{I_D R_1 - V_o}{R_1 + (\beta+1) R_E} = \frac{0.5 \cdot 2.2 - 0.7}{2.2 + (85+1) 0.022} = 0.0978\text{ mA}$$

$\leftarrow I_E = I_B + I_C$

$$I_C = 8.3\text{ mA} \quad V_{CE} = 12 - 8.3 \cdot 1 - 8.4 \cdot 0.022 = \underline{3.5\text{ V}} > 0.2\text{ V}$$



$$r_{be} = \frac{25.9}{0.0978} = 264\text{ }\Omega$$

$$v_o = -\beta i_b R_C$$

$$v_{in} = R_1 i_b + R_E (\beta+1) i_b$$

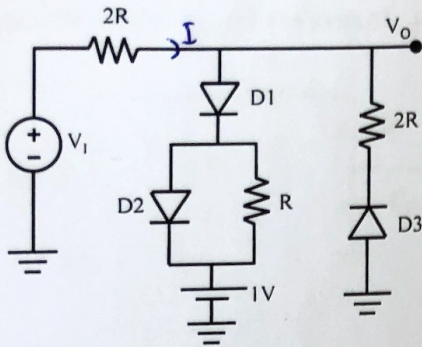
$$i_b = \frac{v_{in}}{R_1 + (\beta+1) R_E + r_{be}}$$

$$\frac{v_o}{v_{in}} = -\frac{\beta R_C}{R_1 + (\beta+1) R_E + r_{be}} = \frac{85 \cdot 1}{2.2 + (85+1) 0.022 + 0.264} = \underline{-19.5}$$

slope	transition points
D3 ON	0.7
all OFF	0.4
D1 ON	0.8
D1, D2 ON	0.6

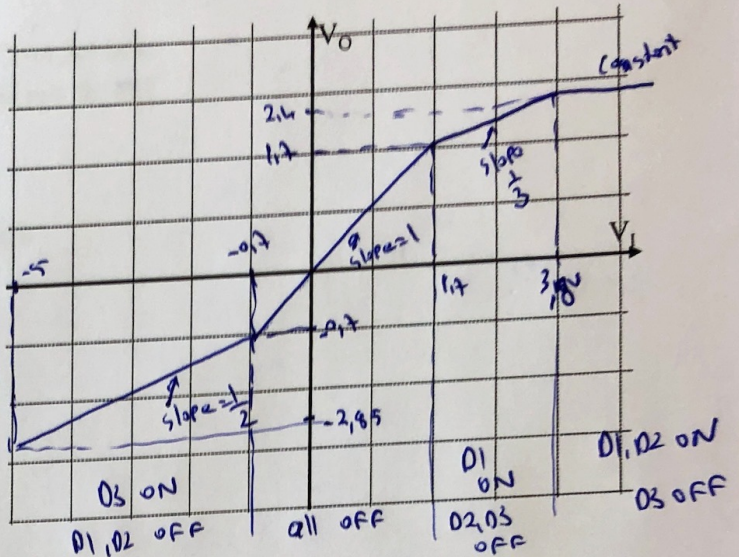
Question 3 [25 Points]:

For the circuit show below, find and plot the output-input voltage relation (V_O vs. V_I) on the provided graph for $-5 < V_I < 5V$. Assume all the diodes have a turn on voltage of $V_{ON}=0.7V$. Clearly indicate all the transition points and the slopes.

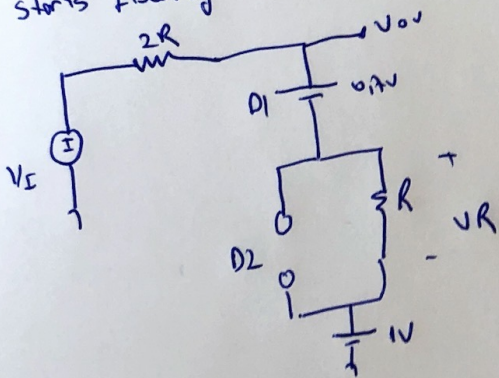


D1, D2, D3 are off

when $-0.7V < V_I < 1.7V$ so $I=0$
and $V_I = V_O$



on the (+) side first D1 turns on (D2 is off) when $V_I > 1.7V$ and the current starts flowing through R



$$\frac{V_O - V_I}{2R} + \frac{V_O - 1.7}{R} = 0$$

$$V_O - V_I + 2V_O - 3.4 = 0$$

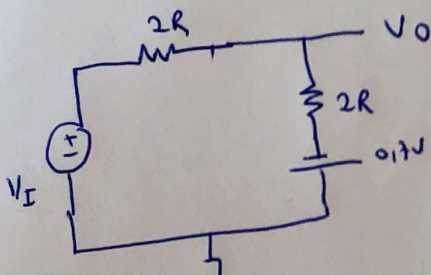
$$V_I = 3V_O - 3.4 \rightarrow V_O, V_I \text{ slope of } \frac{1}{3}$$

when $V_R = 0.7V$ D2 turns on.

$$V_R = (V_I - 1.7) \frac{R}{R+2R} = \frac{V_I - 1.7}{3} = 0.7$$

when $V_I = 3.8V$ D2 turns on and $V_O = 2.4V$ constant for $V_I > 3.8V$

on the (-) side D3 turns on when $V_I < -0.7V$ (D1, D2 off)



$$\frac{V_O - V_I}{2R} + \frac{V_O + 0.7}{2R} = 0$$

$$V_I = 2V_O + 0.7 \quad (V_O, V_I) \text{ slope of } \frac{1}{2}$$

$$-5 = 2V_O + 0.7$$

$$2V_O = -5.7$$

$$V_O = -2.85V$$

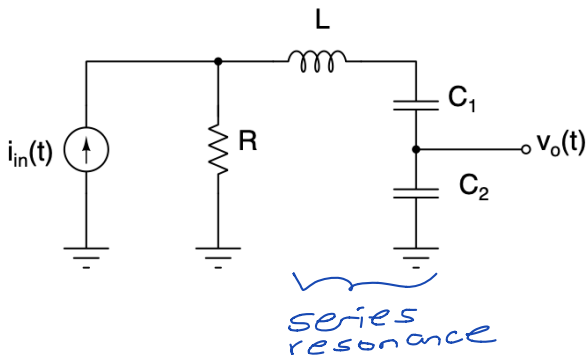
Question 4 [25 Points]:

For the circuit shown below, $R = 1 \text{ k}\Omega$, $L = 2 \text{ }\mu\text{H}$, $C_1 = 3 \text{ pF}$, and $C_2 = 6 \text{ pF}$.

(a) What is the resonance frequency, ω_0 , in units of rps? What is the resonance frequency, f_0 , in units of Hz?

(b) Find $v_o(t)$ if $i_{in}(t) = 6 \sin(\omega_0 t + \pi/6) \text{ mA}$ for all t .

(c) Find $v_o(t)$ if $i_{in}(t) = 6 \sin(10^9 t + \pi/6) \text{ mA}$ for all t .

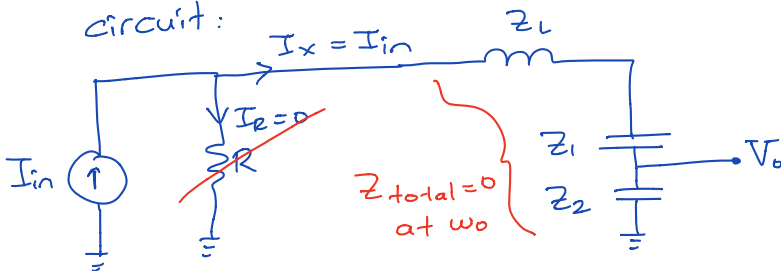


$$a) C_{eff} = \frac{C_1 C_2}{C_1 + C_2} = 2 \text{ pF} \quad (C_1 \& C_2 \text{ in series})$$

$$\omega_0 = \frac{1}{\sqrt{L C_{eff}}} = \frac{1}{\sqrt{2 \cdot 10^{-6} \cdot 2 \cdot 10^{-12}}} = 0.5 \times 10^9 \text{ rps}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{0.5 \times 10^9}{2\pi} = 79.58 \text{ MHz}$$

b) At resonance freq, L, C_1, C_2 together act like short circuit:



$$I_x = I_{in}$$

$$I_R = 0$$

$$V_o = Z_2 \cdot I_{in}$$

$$i_{in}(t) = 6 \sin(\omega_0 t + \pi/6) = 6 \cos(\omega_0 t - \pi/2 + \pi/6) = 6 \cos(\omega_0 t - \pi/3) \text{ mA}$$

$$I_{in} = 6 e^{-j\pi/3} \text{ mA}$$

$$Z_2 = \frac{1}{j 0.5 \times 10^9 \times 6 \times 10^{-12}} = -j \cdot \frac{1}{3} \text{ k}\Omega$$

$$V_o = 6 e^{-j\pi/3} \cdot 10^{-3} \cdot (-j \cdot \frac{1}{3} \cdot 10^3) = -2j e^{-j\pi/3} = 2 e^{-j\pi/2} \cdot e^{-j\pi/3} = 2 e^{-j5\pi/6}$$

$$V_o(t) = 2 \cos(\omega_0 t - \underbrace{5\pi/6}_{150^\circ}) \text{ V}$$

c) At $\omega = 10^9 \text{ rps}$,

$$Z_L = j\omega L = j 10^9 \cdot 2 \cdot 10^{-6} = j 2 \text{ k}\Omega$$

$$Z_1 = \frac{1}{j\omega C_1} = \frac{1}{j \cdot 10^9 \cdot 3 \cdot 10^{-12}} = -j \frac{1}{3} \text{ k}\Omega$$

$$Z_2 = \frac{1}{j\omega C_2} = \frac{1}{j \cdot 10^9 \cdot 6 \cdot 10^{-12}} = -j \frac{1}{6} \text{ k}\Omega$$

$$I_{in} = 6 e^{-j\pi/3} \text{ mA}$$

$$I_x = I_{in} \cdot \frac{R}{R + Z_L + Z_1 + Z_2} \quad (\text{current divider})$$

$$= 6 \cdot e^{-j\pi/3} \cdot \frac{1}{1 + j2 - j\frac{1}{3} - j\frac{1}{6}} = 6 \cdot e^{-j\pi/3} \cdot \frac{1}{1 + j\frac{3}{2}} = \frac{12 e^{-j\pi/3}}{2 + j3}$$

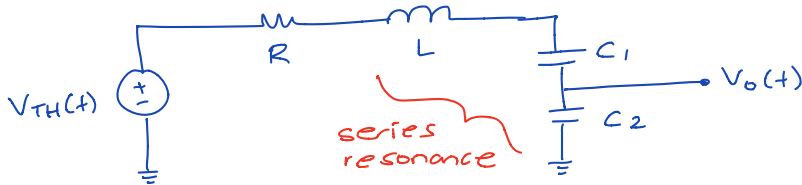
$$= \frac{12 \cdot e^{-j\pi/3}}{\sqrt{13} \cdot e^{j\arctan(3/2)}} = \frac{12}{\sqrt{13}} e^{-j(\pi/3 + \arctan(3/2))} \text{ mA}$$

$$V_o = I_x \cdot Z_2 = \frac{12}{\sqrt{13}} e^{-j(\pi/3 + \arctan(3/2))} \times 10^{-3} \cdot \left(-j \cdot \frac{1}{6} 10^3\right)$$

$$= \frac{2}{\sqrt{13}} \cdot e^{-j(\pi/3 + \arctan(3/2) + \pi/2)} = \frac{2}{\sqrt{13}} e^{-j(5\pi/6 + \arctan(3/2))}$$

$$V_o(t) = \frac{2}{\sqrt{13}} \cos(10^9 t - \underbrace{5\pi/6}_{150^\circ} - \underbrace{\arctan(3/2)}_{56.3^\circ}) \text{ V} = 0.55 \cos(10^9 t - \underbrace{3.6}_{206.3^\circ}) \text{ V}$$

Alternative solution: Use Thevenin equivalent circuit



$$V_{TH}(t) = i_{in}(t) \cdot R$$

a) same as above

$$b) V_{TH} = 6 e^{-j\pi/3} \text{ V}$$

$$V_o = \frac{\frac{1}{j\omega C_2}}{R + j\omega L + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_2}} \cdot V_{TH} \quad (\text{voltage divider})$$

$$= 6 e^{-j\pi/3} \cdot \frac{1}{10^3} \cdot \frac{1}{j 0.5 \times 10^{-9} \cdot 6 \times 10^{-12}} = -j 6 \cdot e^{-j\pi/3} = 6 \cdot e^{-j5\pi/6}$$

$$V_o(t) = 6 \cos(\omega_0 t - 5\pi/6) \text{ V}$$

$$c) V_{TH} = 6 e^{-j\pi/3} \text{ V}$$

$$V_o = \frac{\frac{1}{j\omega C_2}}{R + j\omega L + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_2}} V_{TH} = \frac{-j\frac{1}{6}}{1 + j2 - j\frac{1}{3} - j\frac{1}{6}} 6 e^{-j\pi/3}$$

$$= \frac{e^{-j\pi/2} \cdot 2 \cdot e^{-j\pi/3}}{2 + 3j} = \frac{2}{\sqrt{13}} e^{-j(5\pi/6 + \arctan(3/2))}$$

$$V_o(t) = \frac{2}{\sqrt{13}} \cdot \cos(10^9 t - 5\pi/6 - \arctan(3/2)) \text{ V}$$